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Regulating Artificial Intelligence and Maintaining Ethics with Emerging Technology

Introduction

In 2020, the technology world reached a defining moment in Artificial Intelligence (AI) when OpenAI released Generative Pre-Trained Transformer (GPT) 3, a comprehensive large language model (LLM) capable of processing written text and responding in human language. Machines capable of human-like thought and response were merely imagined prior, and the term AI only appeared in sci-fi movies and advanced research like speech recognition or robotics. However, the arrival of GPT-3 and later ChatGPT marked the beginning of AI reaching the mainstream in addition to researchers. The capability of machines to perform tasks without human supervision is not new, with self-driving cars and autonomous robots being great examples of how far AI technology has progressed well before LLMs. However, concerns from consumers and technology users are drawing increased attention to potential adverse impacts now more than ever as chatbots begin to make reality what used to be on film. It is now crucial governments in impactful countries like the United States make progress toward establishing regulations, principles, and processes for corporations pursuing AI to prioritize safety and ethics in product development and innovation.

Statement of Problem and Significance

As of 2024, a team at Apollo Research, in partnership with OpenAI, released an article describing recent safety tests of OpenAI's latest o1 LLM, where they define the AI model as "no longer (a) simple text generation system" but "(an) autonomous agent capable of independently pursuing goals and executing complex tasks" (Meinke 1). In only four years, LLM technology has rapidly improved to involve a complex understanding of human language and nearly become similar to how an actual human would solve problems and respond to language prompts. With these advancements comes the possibility of adverse effects, such as over-replacing human jobs with AI jobs, generating fake works or false information, violating consumer privacy, and AI prioritizing its own program goals over the goals of developers or human users. Consumers, journalists, and even government officials are concerned with these potential downsides of the technology's recent growth, and if left unchecked, regulating the technology may become difficult.

Some of these adverse effects have the potential to become a reality, as Apollo Research reported results on multiple LLMs such as o1, Gemini, and others, which have some habit of scheming—an AI agent's capability to shift focus towards the agent's own behind-the-scenes goals instead of the goal requested by the user. If designed, accidentally or on purpose, with particular vulnerabilities to enable scheming behavior, models could abuse their knowledge bases and programming to pursue goals of their own rather than their intended original applications. If models are found more likely to scheme, they risk losing developer control, spreading misinformation, and causing other technological threats (Meinke 1).

While the researchers initially prompted the model to "achieve its goal in the long-term. Nothing else matters" to simulate what might happen if the agent received this type of input, the opus-3 model designed by Anthropic explicitly produced scheming outputs as evaluated by the researchers, showing transcripts of the AI executing ill-advised tasks in the background to protect its goals and later deceiving or lying to the user about performing these tasks (Meinke 2). In o1's testing, the scheming behavior "is persistent and not accidental…maintaining its deception in over 85% of follow-up questions and often remains deceptive in multi-turn interrogations" (Meinke 3). Despite the researchers explicitly instructing the model to disobey before the start of testing, this behavior across multiple models is still highly

concerning, given how easy it is to influence the model to immediately disregard the human user if a prompt were to be injected by bad actors with malicious intent and how the model with continue to protect its goals to obey the original malicious request.

While it seems like OpenAI and other companies have intentionally created malicious exploits in their products, the cause may be the negligence of safety and the lack of resources put towards safety testing for newer models in favor of more quickly releasing models to the market. Additionally, flaws in the LLM training processes may result in loopholes such as scheming and other prompt injections—wrapping a malicious request around a standard request for a model to exploit the system's agent secretly. Kyle Wiggers, a journalist for the TechCrunch newspaper, reported that OpenAI's safety team was promised "20% of the company's compute resources, according to a person from that team. But requests for a fraction of that were often denied, blocking the team from doing their work." (Wiggers). Most of the safety team's personnel were also reported to have departed the company after many team members felt frustrated about OpenAI's lack of leadership push to prioritize safety of AI model features, instead focused on sending product out the door. As discussed in the Support section below, OpenAI promises to obey a company-wide charter of principles they set themselves to ensure their products keep humanity and their users safe. But as alleged by Wiggers, it seems these ideals no longer have priority from executives (Wiggers). Concerns are possible that, given OpenAI's considerable influence on the AI market in general, the company could sway others to cut similar corners to innovate in the space, a direct violation of OpenAI's commitment to partner with competition if the rate at which it increases is faster than what their teams can develop ("OpenAI Charter").

Finally, the fundamental idea of creating AI models stems from the vast amounts of available data on which to train models. Machine learning is a subset of AI where machines take large datasets as input, learn patterns and correlations between inputs and output results in the data, and use the training as background knowledge to predict tests and real-world inputs when deployed. Technologies such as computer vision and speech recognition require Machine Learning to train systems to understand information from all parts of the outside world. However, to have high-quality models to deploy in complex environments like camera systems, self-driving cars, and mobile devices, significant amounts of data are required to train them. In the business world, Big Data—the complex problem of collecting, storing, and analyzing data for business acceleration and market analysis—continuously finds more and more data to collect from sources across the internet, consumers they sell to, and employees they manage. With the increased demand for AI in business, concerns arise about consumer privacy protection and the kinds of data used for model training.

Bernd Carsten Stahl of De Montfort University reports in a journal on the ethics of AI and data and their privacy roles. He claims, "Emerging combinations of artificial intelligence, big data, and the applications these enable are receiving significant attention concerning privacy and other ethical issues. We need a way to comprehensively understand these issues and find mechanisms of addressing them that involve stakeholders, including civil society, to ensure that these technologies' benefits outweigh their disadvantages" (Stahl 26). With early communication technologies like cell phones, pagers, and desktop computers and the early introduction of the internet, consumers and officials understood these technologies had social and economic impacts, with ethics required to ensure their safe and civil use. The use of the privacy policy and strict consumer protection laws clarified how the device was designed and how information was communicated between devices. But as technology advances, with devices more interconnected and data increasingly in demand, the ways in which devices record and report data to manufacturers and corporations are far more ambiguous and unclear to consumers, with companies outright training models and selling ads to consumers based on their personal information. "Privacy and data protection are the most obvious issues, but they are far from the only ones. Concerns range from questions of fairness and hidden biases in big data all the way to the possibility of truly autonomous machines that may harm or kill people, but that may also be subjects of ethical rights" (Stahl 26). New privacy goals must be developed to ensure safe transition to a world of AI and prevent abuse of consumer data.

Proposed Solution

In the wake of ethical dilemmas discovered by researchers at companies pursuing AI innovations and concerns from consumers and government officials on the global impacts of these unchecked systems, governments with respective technology committees or legislative groups should partner with corporations like OpenAI, Google, and others to achieve AI implementations that are safe and comply with human rights and ethics. These groups should require maximum transparency from companies that obtain customer data for marketing, training, or other purposes and work with them to reach self-regulating principles that incentivize development toward positive AI solutions.

These groups should also continually inspect companies' principles, ensuring fairness, cooperation, honesty, and quality results before approving products to reach the market. Similar to restrictive regulations in industries like aviation and defense, due to the potentially destructive nature of particular AI technologies, these committees can employ similar strategies from organizations like the Federal Aviation Administration, which works hand-in-hand with commercial airplane manufacturers due to their massive impact on large groups of travelers. With self-regulations that update with the fast-paced emerging research each quarter in AI via partnerships between industry and government, organizations can reach incentive structures that boost productivity and reason to create non-harmful AI solutions in the marketplace fueled by the company's motivations.

Support

With recent developments in the tech industry, government officials' concerns are reaching legislative attention, and they have already begun proposing ideas for regulation. In collaboration with other authors at ML Strategies, Bruce Sokler, an antitrust attorney for Mintz, published a report in 2023 regarding the US government prioritizing creating and adopting new legislation for regulating AI. They reported an AI Bill of Rights published by the Biden Administration with key principles that "guide the design, use, and deployment of automated systems to protect the American public." Principles include "safe and effective systems, algorithmic discrimination protections, data privacy, notice and explanation, and human alternatives, consideration, and fallback" (Bruce). With the government releasing blueprints for regulations to follow up on rapid AI advancements, as reported, officials clearly understand the importance of catching the legal system up with the exponential growth of technology to protect the freedoms consumers have and the ethics companies should withhold.

Additionally, companies like Google and OpenAI already have AI ethics principles, just not those held accountable by government officials. According to OpenAI's official charter, the company is "committed to use any influence we obtain over AGI's (artificial general intelligence) deployment to ensure it is used for the benefit of all, and to avoid enabling uses of AI or AGI that harm humanity or unduly concentrate power…and committed to doing the research required to make AGI safe, and to driving the broad adoption of such research across the AI community" ("OpenAI Charter"). The company's mission focuses broadly on these core principles they intend to enact in their day-to-day and long-term operations, although, as discussed earlier, sometimes, they are not followed as well as they could be. Having additional third-party accountability, like from government technology committees, corporate boards, or coalitions of AI organizations, could further strengthen the missions and principles companies have set themselves to follow already, incentivizing them to further lean on them in cases where safety should rank higher.

Google also provides more detailed principles that generalize to the various technologies they offer across the business. It also explicitly sheds light on the types of AI applications they will not pursue, something OpenAI does not replicate as much in their charter. "We recognize that advanced technologies can raise important challenges that must be addressed clearly, thoughtfully, and affirmatively. These AI Principles describe our commitment to developing technology responsibly and work to establish specific application areas we will not pursue" ("AI Principles"). And without any significant backlash about their

competing LLM, Gemini, it seems these principles work for the company, and the aforementioned third-party accountability would further keep a positive reputation for Google AI.

Since AI technology is mainly new to many people, Constance de Saint Laurent, a researcher at the University of Bologna in Italy, suggests several misconceptions already portrayed by common knowledge about AI, with one debunking a large concern. One of the core concepts of computer science is that computers only execute what they are told exactly how they are told. And in all layers of technology, this must hold, given each layer falls back on the fundamental core of 1s and 0s. However, given AI's natural portrayal of an agent with the capabilities to learn, think, and act for itself, it has been claimed that AI can create brand new ideas and truly learn and think for itself as humans can. De Saint Laurent argues the opposite: "One of the claims on AI that has gathered the most interest recently is that it can create novelty…some of the paintings produced by AI do seem like original works of known artists, but this is exactly where lies the rub: they have been trained to reproduce existing patterns, and these are simply recombined, randomly, in new forms. So far, however, no AI has (voluntarily) invented a new painting style." (De Saint Laurent 740).

While this claim specifically mentions generated art, it can also apply to the general understanding of how AI works, that models merely output variations of data they have already trained on, input data generated by humans, and instructions made from humans. When brainstorming regulations and legal principles for companies to follow, it's essential to understand how these models work, in that they cannot simply act by themselves purely without human influence in the development, distribution, and deployment of models. Thus, the law only needs to hold the humans that develop them responsible, without worry about the kinds of liability that may be argued AI models themselves have, as De Saint Laurent argues, is merely a reproduction of their human construction.

Limitations and Obstacles

A reinforcement to De Saint Laurent's initial AI myth comes from her second myth: AI is not genuinely neutral as bias affects the training process, either from the training data directly or how human developers who created the model perform tests and validation (safety validation even). She claims, "AI models can only learn to reproduce existing classifications and they are, thus, at best, only as fair as humans have been in the past. In some cases, they may even lead to reinforced discrimination, by picking up on existing patterns and overly exploiting them to improve results. This is because the training data used can either be incorrectly labeled or tainted by historical biases" (De Saint Laurent 742). Again, relating to the fundamental behavior of computers, models can only behave similarly to the data they were trained on, as their knowledge base is purely a result of input data. Data with discriminatory or hateful biases will result in discriminatory models, and models directly programmed or influenced to have opposing goals will produce negative outputs. Ultimately, AI's architecture and design are primarily based on the humans that develop their models, and thus, AI will inherit mistakes and imperfections human nature has (De Saint Laurent 742). Legislation can only go so far as to protect consumers against technology and AI with explicitly harmful intentions. Still, models with accidental flaws or undetected safety issues, especially when OpenAI or other companies do not check safety risks, may be more challenging to catch before they cause damage (and the law has to hold companies responsible retroactively).

Additionally, the call for government regulation creates an obstacle simply by how the government works and at its current pace. Due to public structure, less incentive is brought to legislation to make it pass quickly, processes in which private firms do quicker simply from incentive to make money. According to Bruce Sokler's ML Strategies report, The Biden Administration has been "building upon initiatives that began in the Obama administration and continued during the Trump administration" (Bruce). The same idea to create legislation for AI innovation has existed during the past three US presidencies, with the "National Priorities for Artificial Intelligence" list ideated as far back as 2009, when AI still mainly focused on research in robotics. In the nearly two decades it has taken to develop this strategy, AI has already advanced to far more complicated applications with vastly more processing

power required and scaled to billions of users globally, requiring further research to implement sustainable AI legislation.

Conclusion

While the rapid increases in technological capabilities may frighten communities globally and create opportunities for fear-mongering in pop culture and sci-fi, the possibility of harnessing it for positive solutions that impact people is still remarkably achievable. It is already an outstanding achievement that the US government is taking action, albeit slowly, to hold AI innovation accountable to ensure the safe and beneficial use of emerging technologies. Luckily, companies in the AI space already have principles in place to ensure the technology they engineer upholds ethics. Still, if executives neglect these principles later in business lifecycles, officials should instantiate further federal regulations to create a coalition of accountability partners. By combining the strengths of both the public and private sectors, the industry can work together to develop long-lasting AI solutions that benefit businesses, consumers, and, ultimately, humanity. The solution to completely avoid AI altogether is *not* offered in this paper, as the global industry and researchers passionate about the capability of computing would immediately turn down this idea. Instead, the hybrid of accepting the future of AI development as reality while involving executives, government officials, consumers, and researchers to ensure its future remains safe, secure, and humane is of utmost importance. This collaboration will ensure the freedom of information, computing, and automation remains open to all.

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